

AI Provider: Hugging Face
(Llama-3.1-8B)

Database Health: HEALTHY

Scope: Real-Time + PySpark Batch

1. Executive Summary & Operational KPIs

Total Volume	Avg Roundtrip Latency	System Quality Score	Poor Defect Rate
37	9.86s	82.3 / 100	35.0%
Production requests	SLA Baseline: < 8.0s	Tier: EXCELLENT / GOOD	Baseline: < 20%

2. Language-Pair Performance Breakdown

Language Pair	Requests	Avg Latency	Quality Score	Observed Quality	Defect Rate
EN → HI	20	8.35s	83.6 / 100	61.5%	38.5%
EN → JA	2	4.07s	92.0 / 100	100.0%	0.0%
HI → EN	7	18.89s	81.7 / 100	75.0%	25.0%
EN → ES	4	3.51s	92.0 / 100	100.0%	0.0%
EN → DE	2	4.59s	60.0 / 100	0.0%	100.0%
EN → BN	2	17.15s	88.0 / 100	100.0%	0.0%

3. Human-in-the-Loop Feedback & Stylistic Preferences

Defect Taxonomy Breakdown			Evaluator Style Preferences		
Defect Category	Flagged Count	Share (%)	Candidate Style	Selections	Preference (%)
Other	2	28.6%	Natural	4	80.0%
Too Literal	4	57.1%	Formal	0	0.0%
Terminology Issue	1	14.3%	Literal	1	20.0%

4. Apache Spark (PySpark) Batch Data Quality Audit

PySpark ETL Metric	Record Count	Operational Definition
Total Historical Input	68,025	Raw ingestion across batch partitions
Clean & Valid Records	66,446	Schema verified, non-null, ready for fine-tuning
Invalid Records Isolated	1,579	Corrupt encodings or malformed language codes
Exact Duplicates Filtered	531	Removed via distributed SHA-256 deduplication
Missing / Empty Outputs	817	Zero-length or truncated translations dropped
Statistical Anomalies	200	Length ratio or latency exceeding 3-sigma

5. Automated Continuous Improvement Recommendations

- *'Too Literal' is currently the most frequently reported defect reason (57.1% of negative feedback).*
- *'Natural' variant demonstrates the highest user selection rate (80.0% of preferred choices).*
- *Operational anomaly detection flagged 7 translation requests requiring review.*